

Complex data types in BigQuery

As a cloud data professional, you'll work with complex data types while using different tools. Throughout this certificate program, you'll work with BigQuery to put some of the skills and knowledge you've developed into practice. As you work more with BigQuery, it's important to be aware of common complex data types. In this reading, you'll explore these common complex data types so that you're prepared to work with them in the future.

Arrays, structs, and JSON

There are three complex data types widely used in BigQuery: arrays, structs, and JSON files. Each of these data types have unique uses and benefits that users can take advantage of to store and analyze data more efficiently.

Array

An array is a collection of values of the same data type. Arrays are one of the most commonly used complex data types in BigQuery. They can be used to store lists of values. Arrays can be nested, meaning you can store arrays within arrays. This is particularly useful for storing hierarchical data. For example, an array of pet types could be stored as:

```
['cat','dog','bird','hamster','turtle'].
```

Data like customer purchase history, employee IDs, or temperatures can also be stored as arrays.

Struct

A struct is a collection of named fields. Structs allow users to group related fields together in BigQuery, and store complex data structures as a single field. Here's an example of a struct that contains information about the price and discount for toothpaste on sale at a retail store:

Unset

```
'sales_items': {  
  "product_type": "toothpaste",  
  "price": "3.75",  
  "discount": "0.50"  
}'
```

Structs can also contain information like customer contact information, students records, or athletic records.

JSON

JSON, or JavaScript Object Notation, is a text-based data-interchange format. JSON data is structured in a key-value pair format that can be nested and used to store complex data structures. These files can contain data like user profiles, product information, or geographical data.

File formats and data types

Some of these data types, like JSON data, are actually file formats. The JSON data enables data to be stored in its natural JSON format. This enables analysts to infer the schema-on-read instead of having to define the schema up-front in tools like BigQuery. This results in more flexible data storage and schemas.

Key takeaways

As a cloud data professional, you have a wide variety of different data and file types you can use to store, access, and analyze. Arrays, structs, and JSON are just a few examples of commonly used data types you can use in BigQuery. As you grow in your skills, there will be lots of exciting ways to leverage these data types. Understanding how they're useful, and what they can do for you, will also enable you to work more efficiently and effectively.